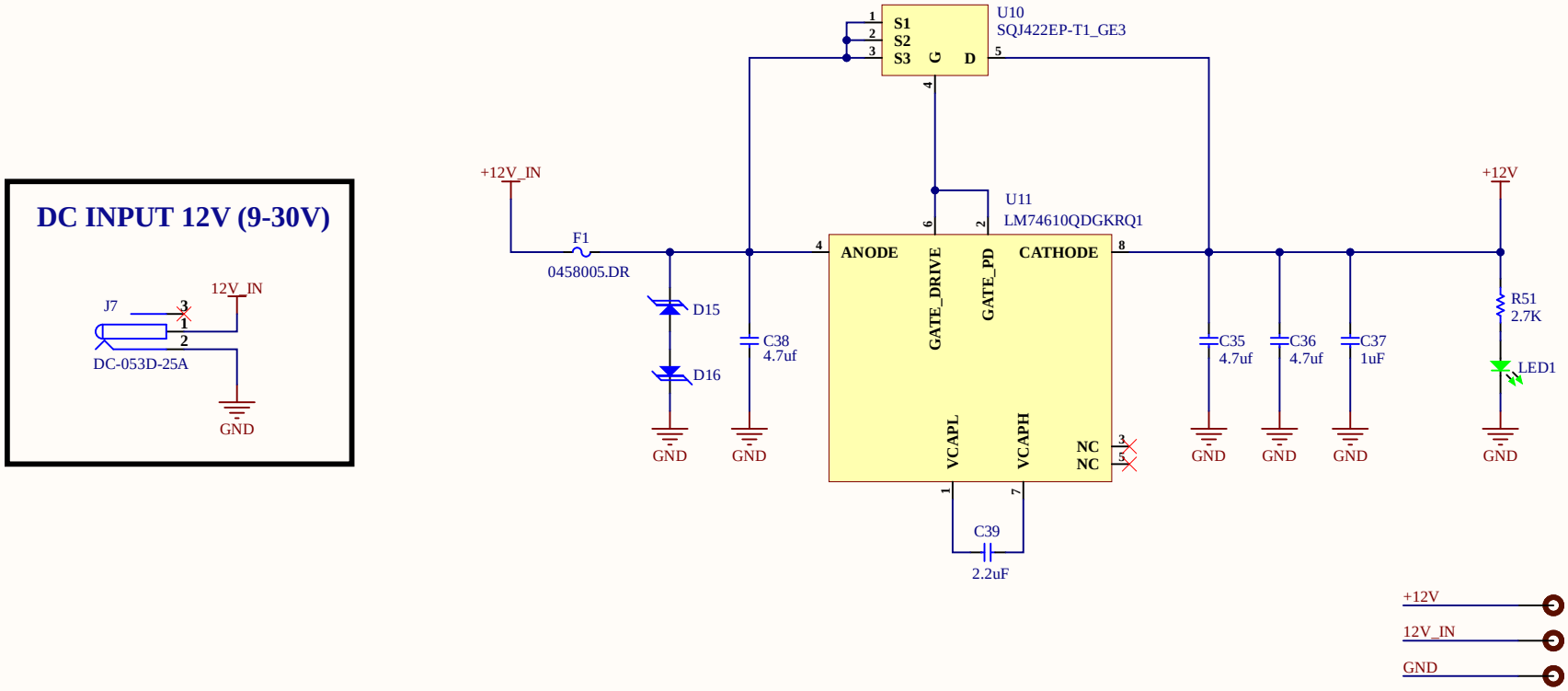
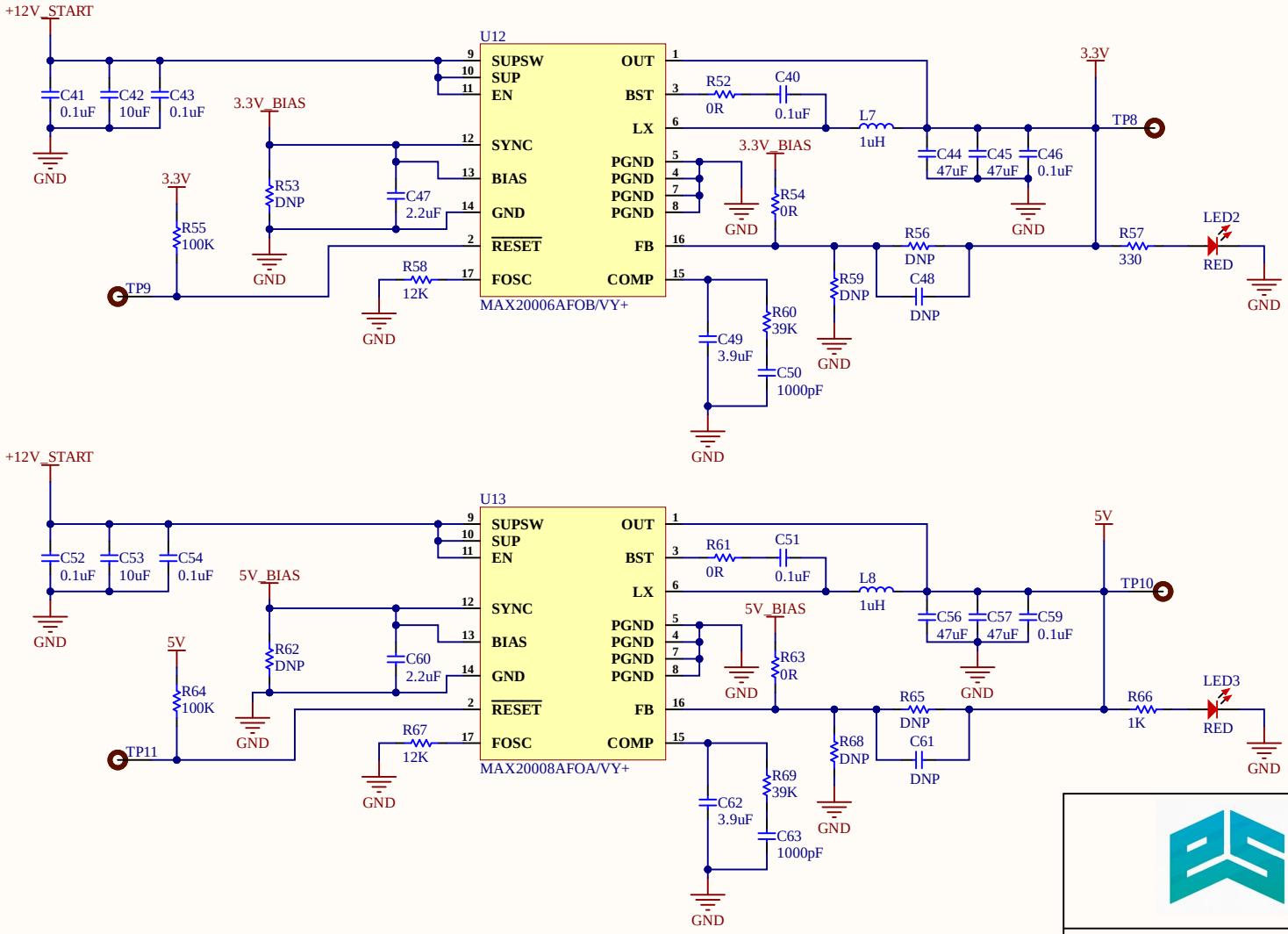
INPUT PROTECTION CIRCUIT



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Date: 15/03/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 16

DC_DC_CONVERTER



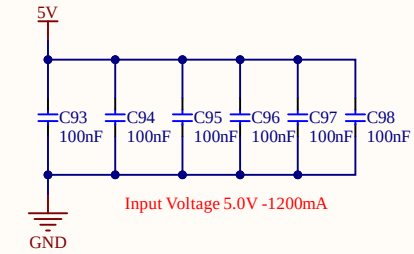
eSpark Consulting Group

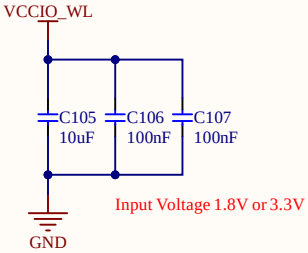
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File: Carrier-Board-v1.0.PrjPcb	
Date: 14/03/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 16

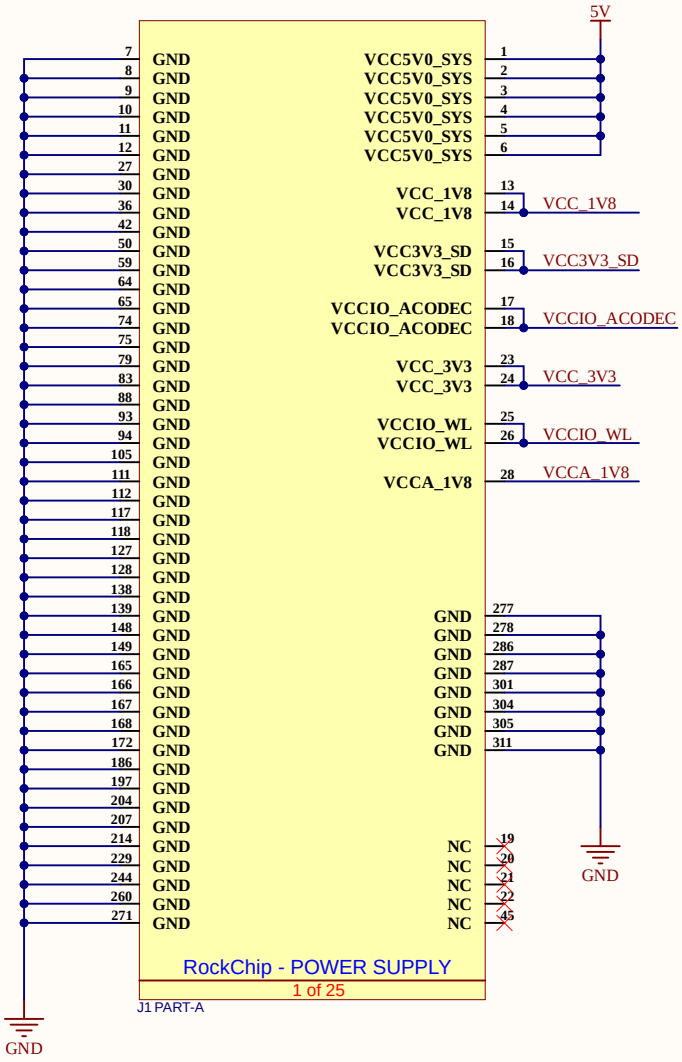
ROCKCHIP POWER SUPPLY

DECOUPLING CAPACITORs

PLACE CLOSE TO EACH PIN

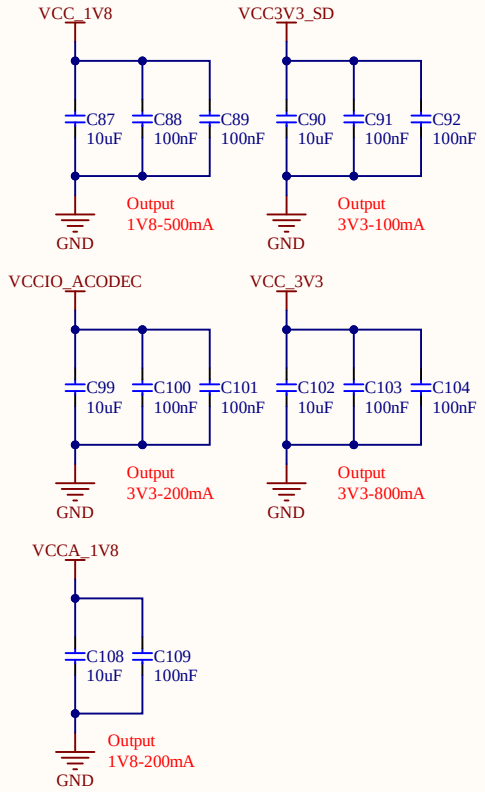






DECOUPLING CAPACITORs

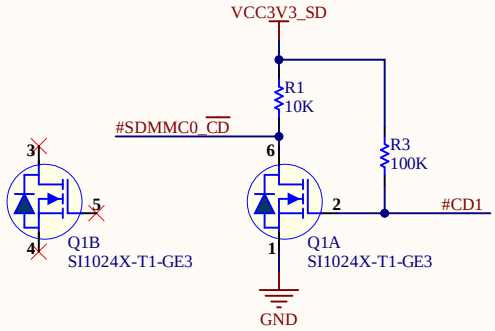
PLACE CLOSE TO EACH PIN

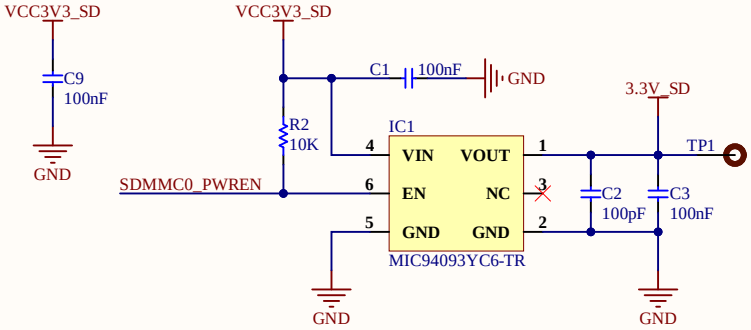


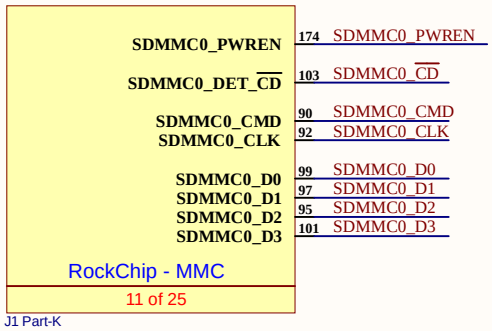
eSpark Consulting Group

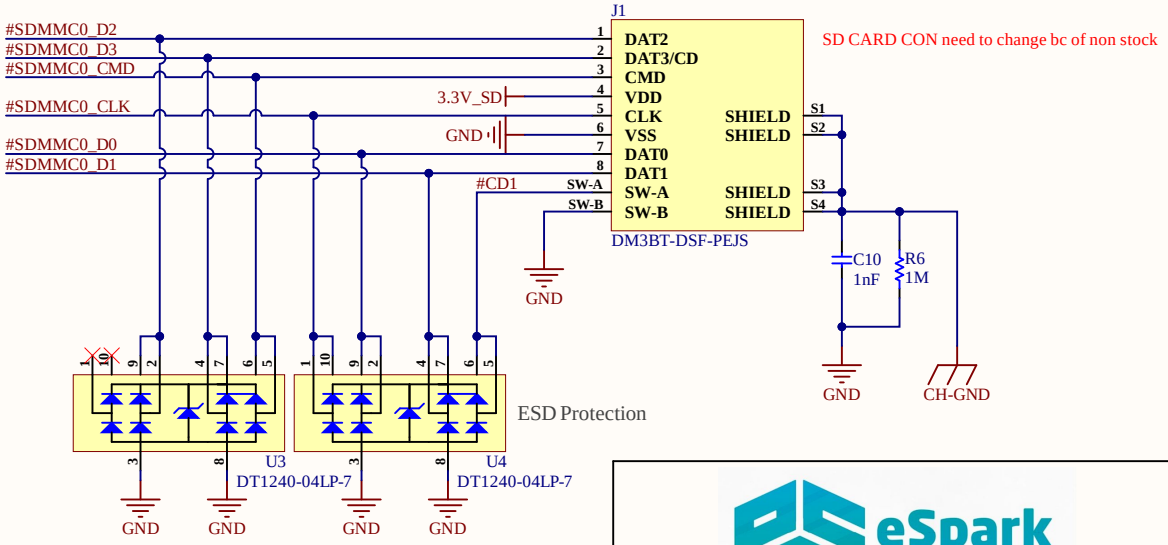
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File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 16

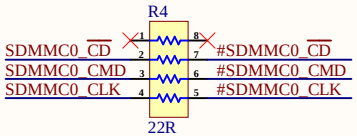
MICRO SD CARD INTERFACE

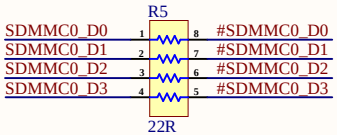








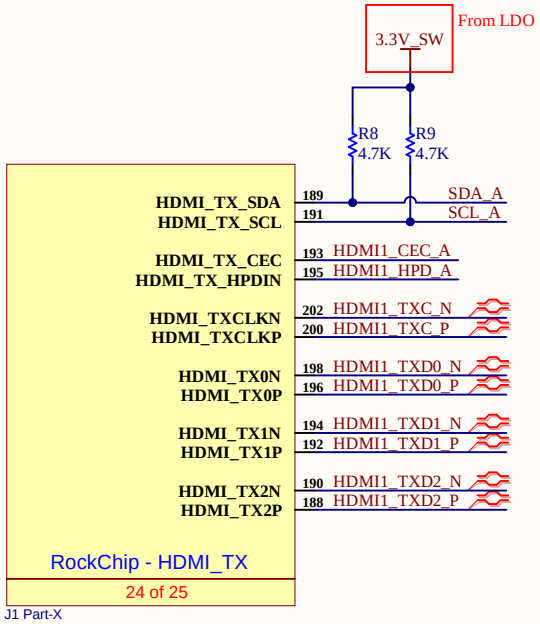


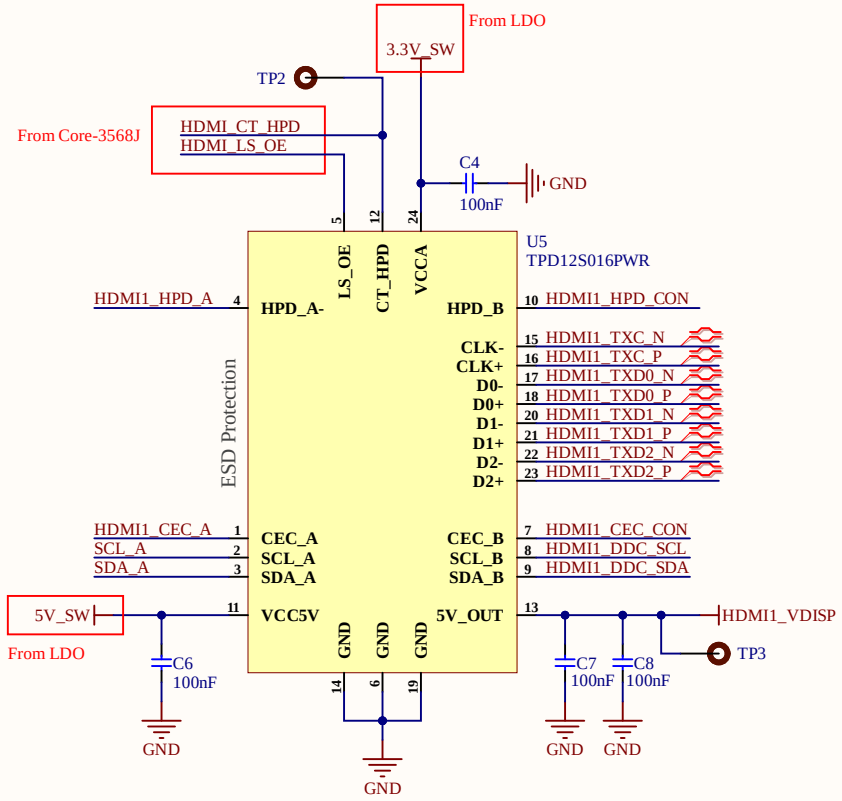


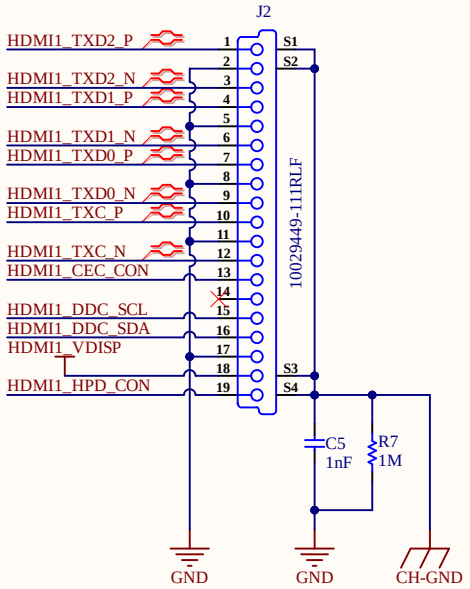
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Title: MICRO_SD_INTERFACE.SchDoc	
File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 16

HDMI INTERFACE

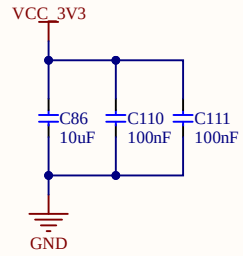


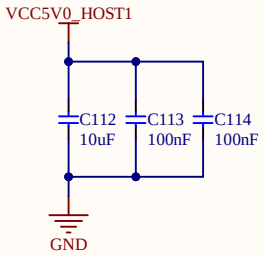


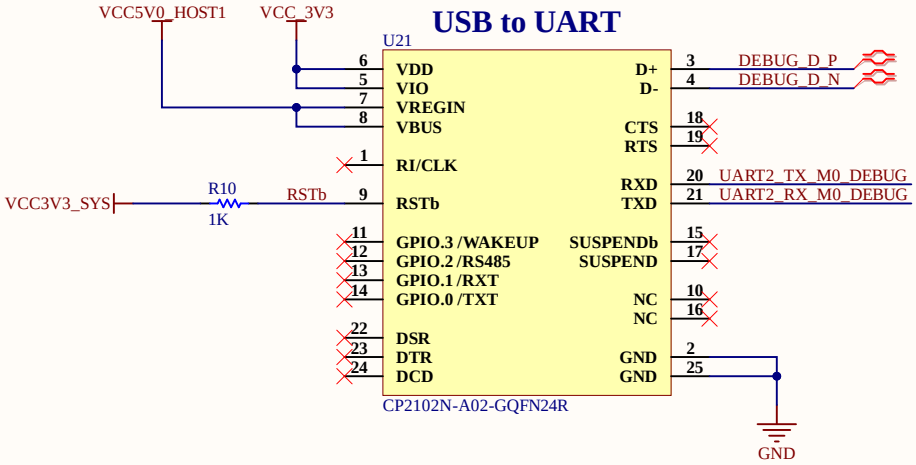


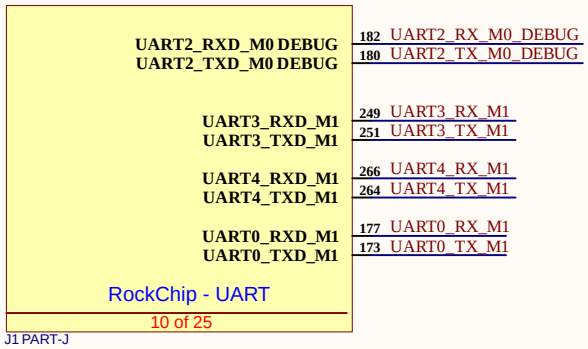
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File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 16

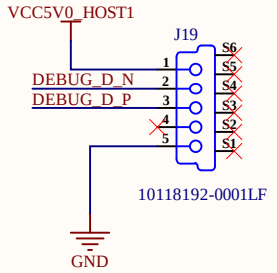
DEBUG_UART_INTERFACE

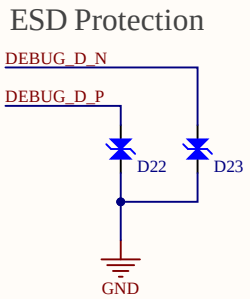


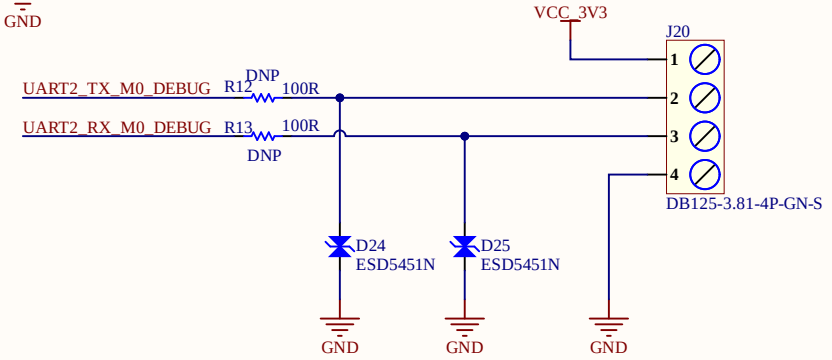










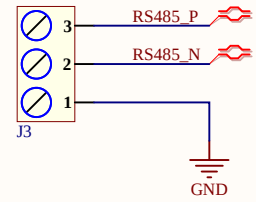


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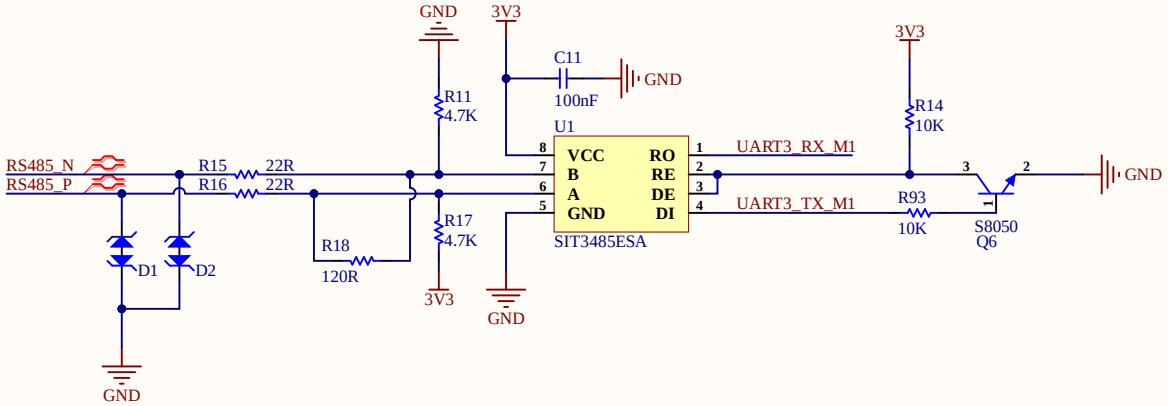
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File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 16

RS232_RS485_INTERFACE

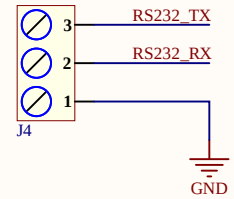
PORT for RS485



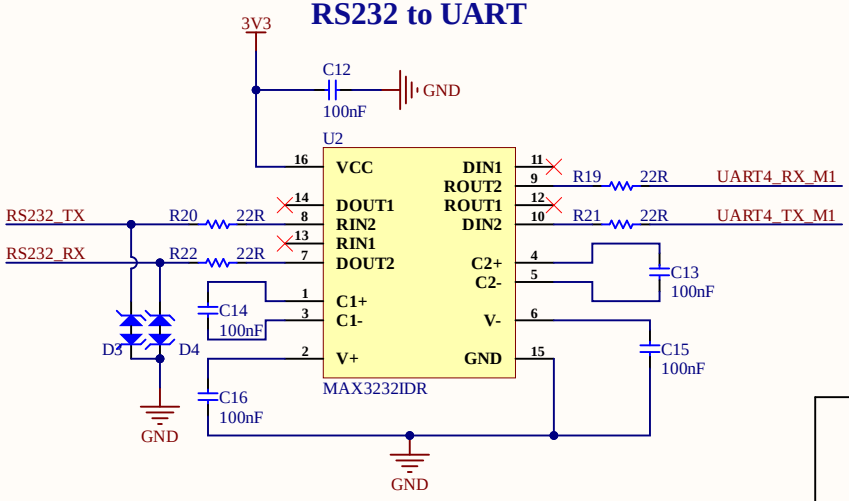
RS485 to UART



PORT for RS232



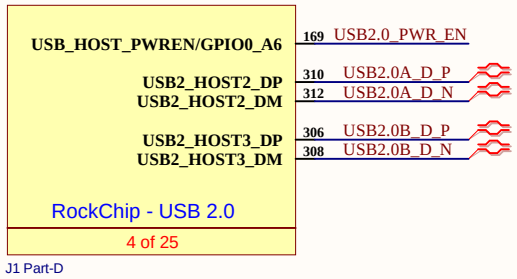
RS232 to UART

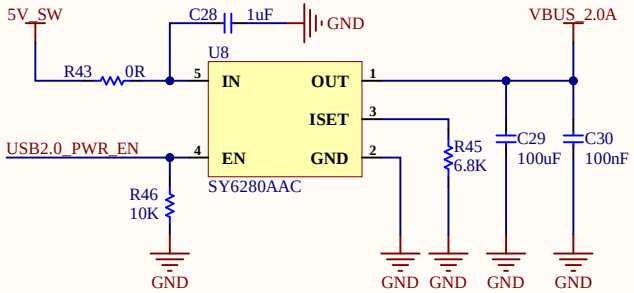


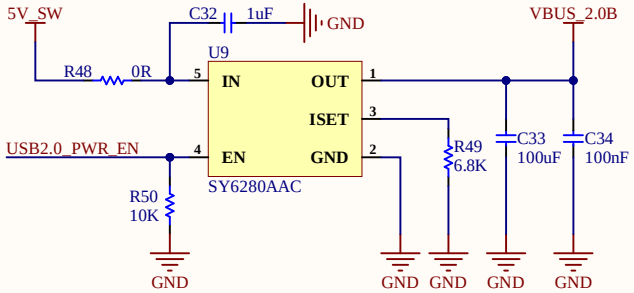
eSpark Consulting Group

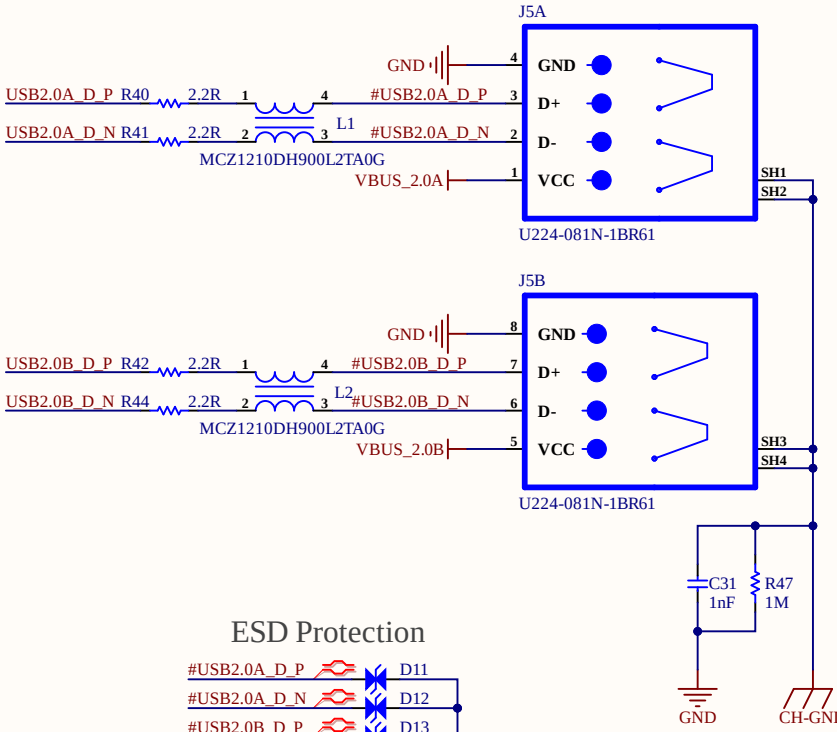
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Date: 15/03/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 16

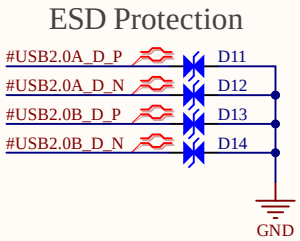
USB2.0 INTERFACE







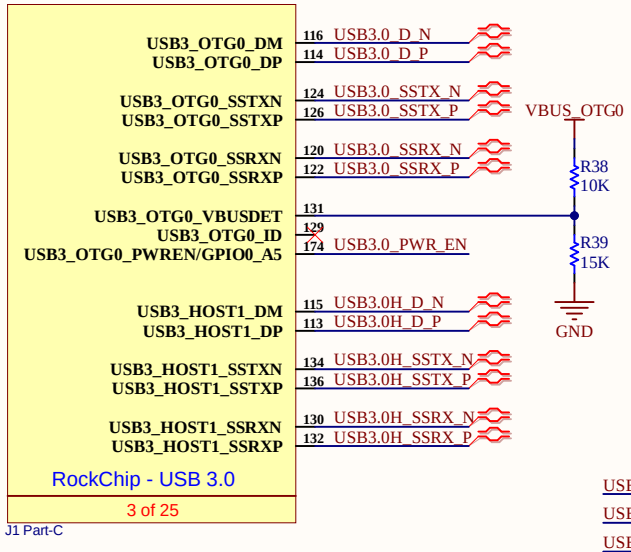




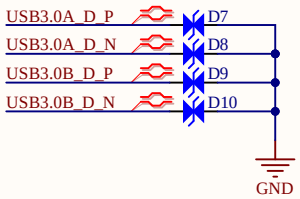
eSpark Consulting Group

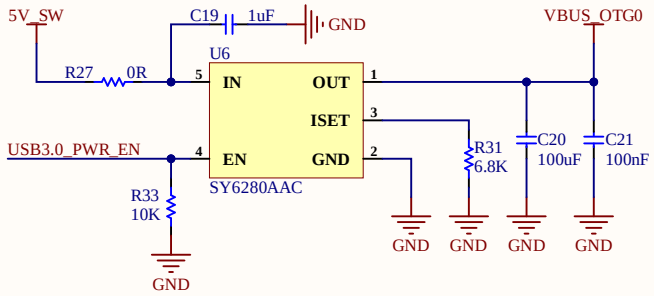
Title: USB2.0_INTERFACE.SchDoc		
File: Carrier-Board-v1.0.PrjPcb		
Date: 14/03/2024	Page	Num: 8
Drawn By: Huzaifa Ali	Page	Total: 16

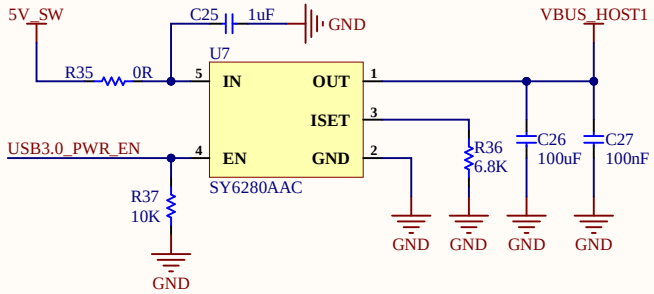
USB3.0 INTERFACE

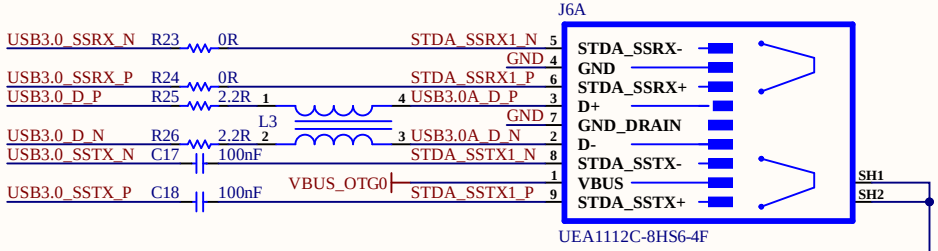


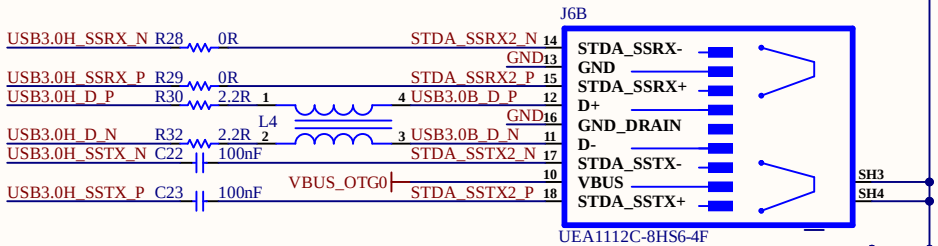
ESD Protection

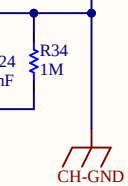




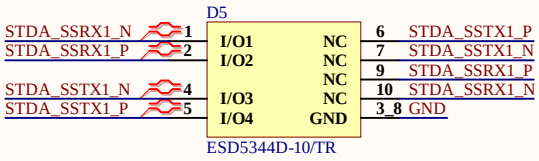


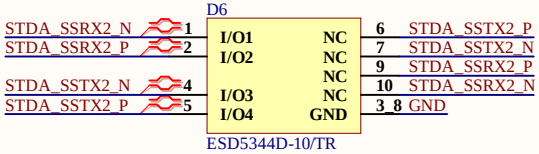


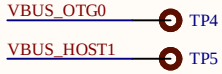




ESD Protection

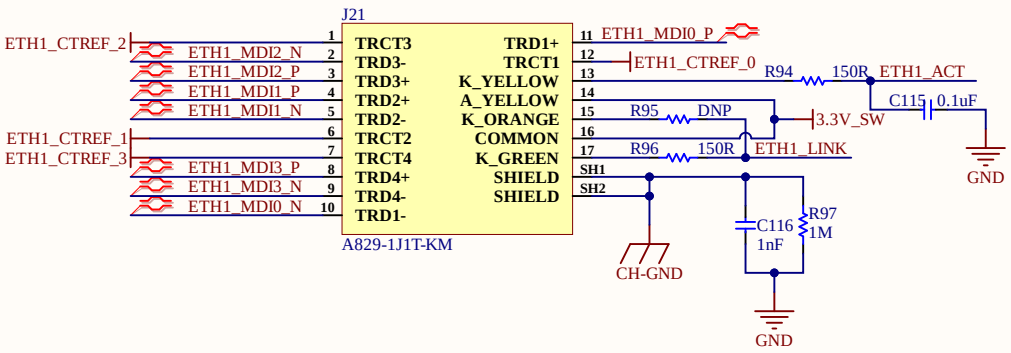


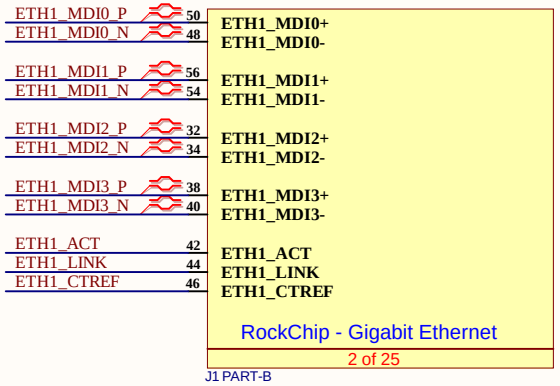


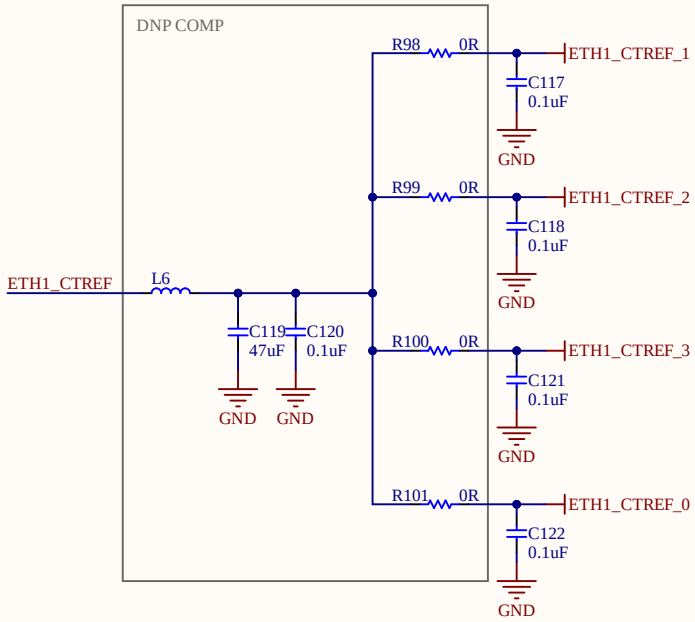


Title: USB3.0_INTERFACE.SchDoc	
File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 16

ETHERNET_PHY0_INTERFACE



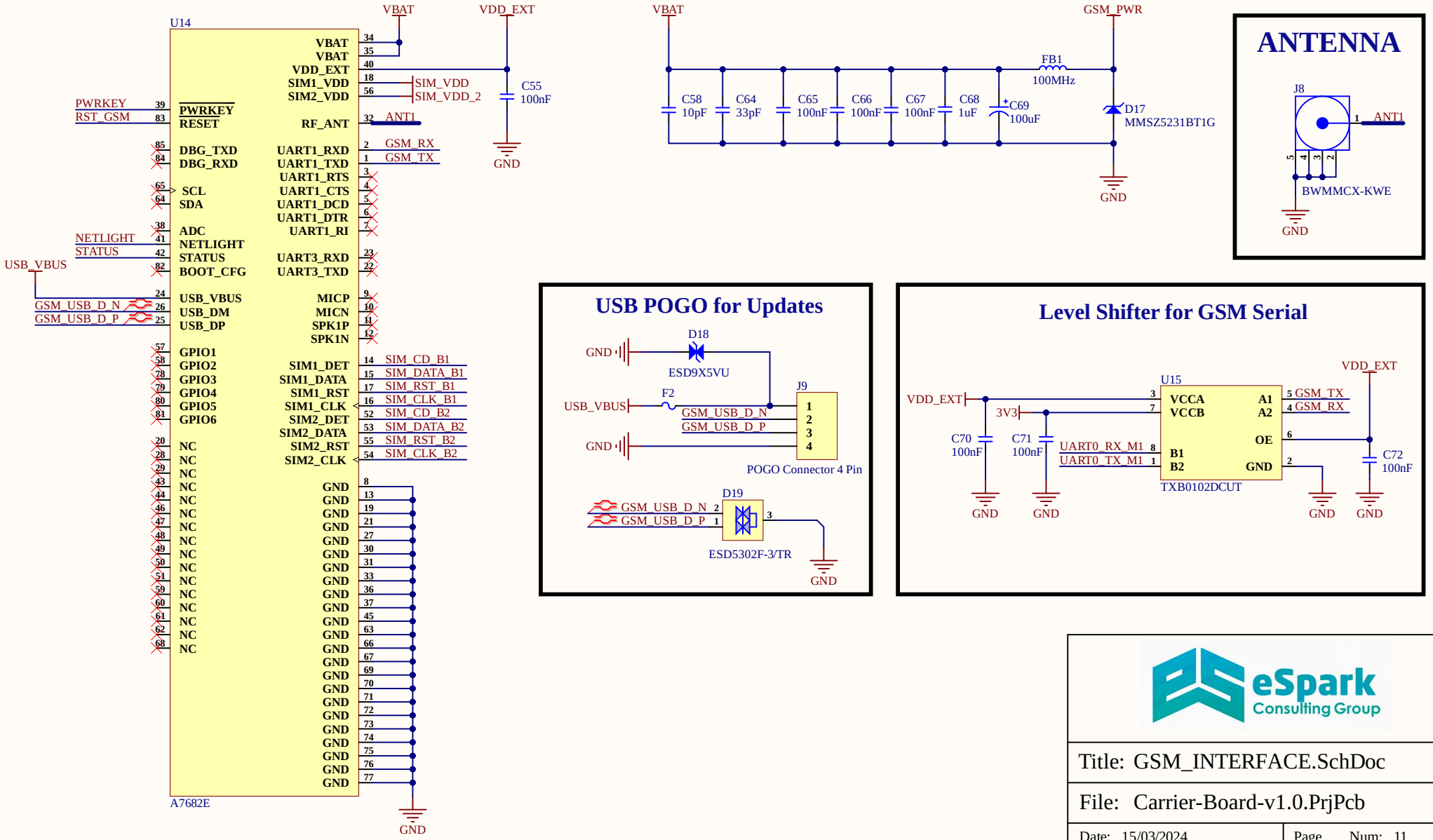




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Title: ETHERNET_PHY0_INTERFACE.SchDoc	
File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 16

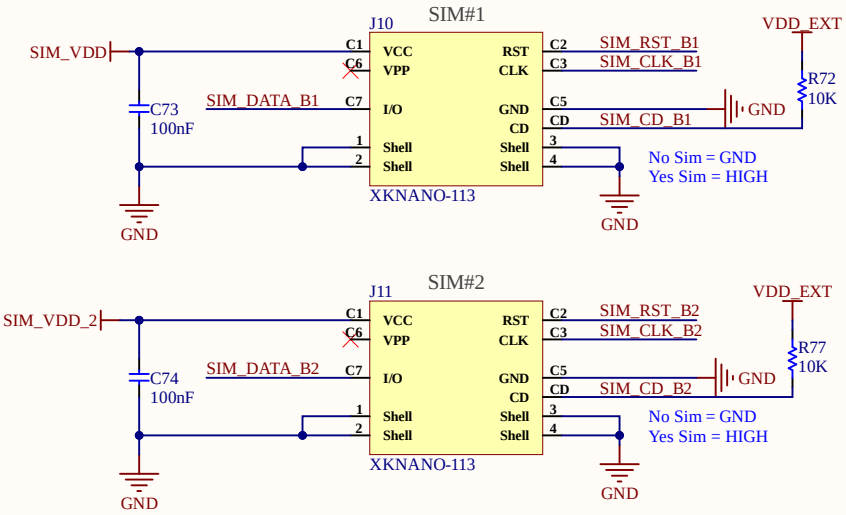
GSM_INTERFACE



Title: GSM_INTERFACE.SchDoc	
File: Carrier-Board-v1.0.PrjPcb	
Date: 15/03/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 16

DUAL_SIM_INTERFACE

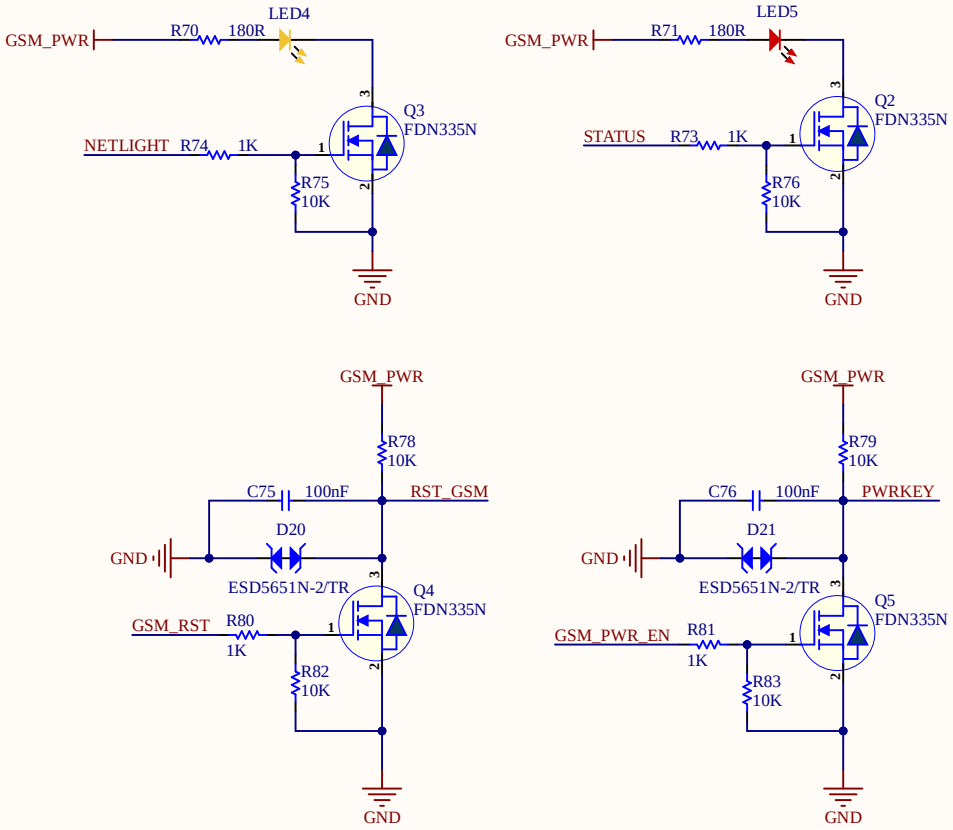
DUALSIM WITH MULTIPLEXER



SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:

- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

STATUS LEDs

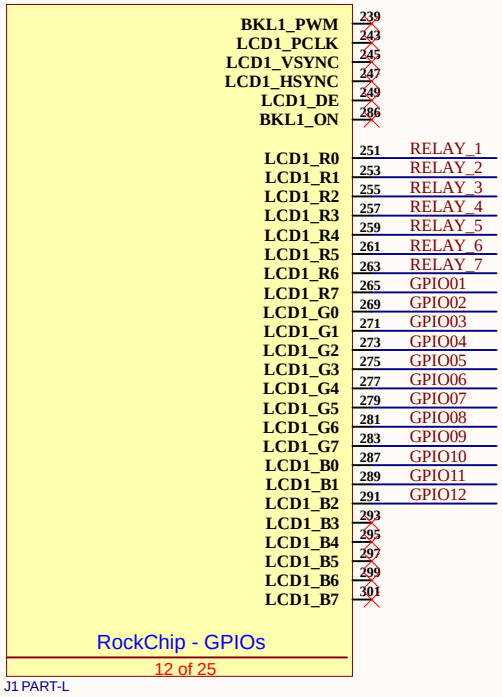


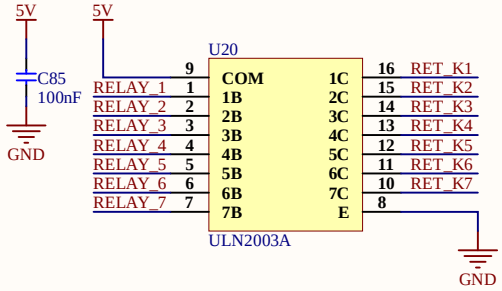
eSpark Consulting Group

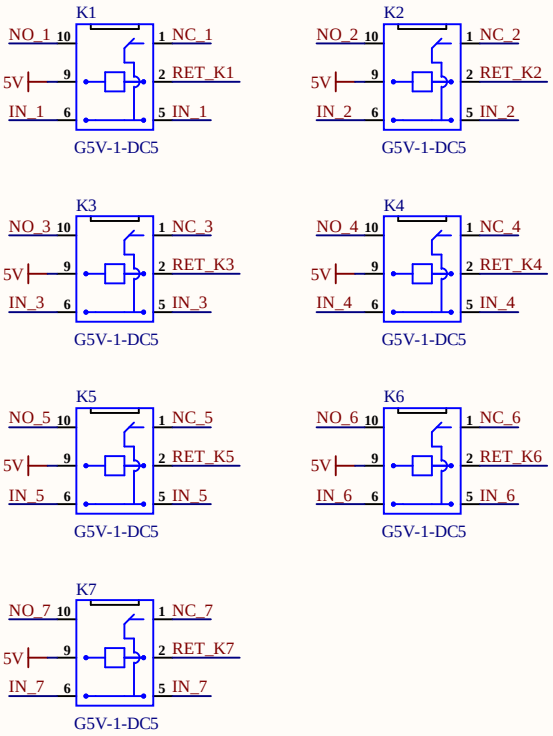
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Date: 14/03/2024	Page	Num: 12
Drawn By: Huzaifa Ali	Page	Total: 16

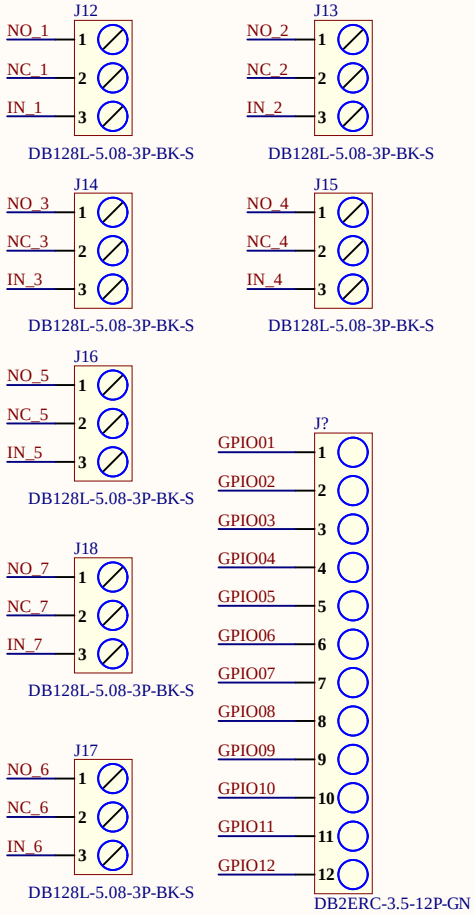
GPIOs_RELAY_SHEET

RELAY WITH DRIVER







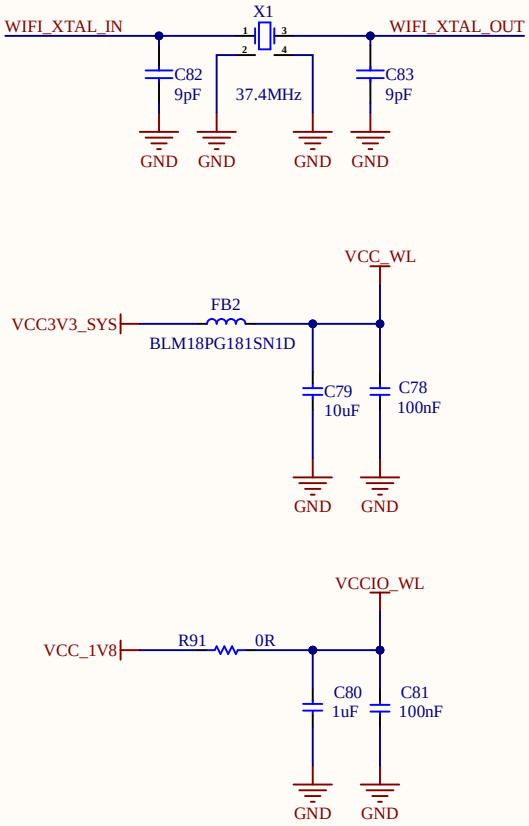


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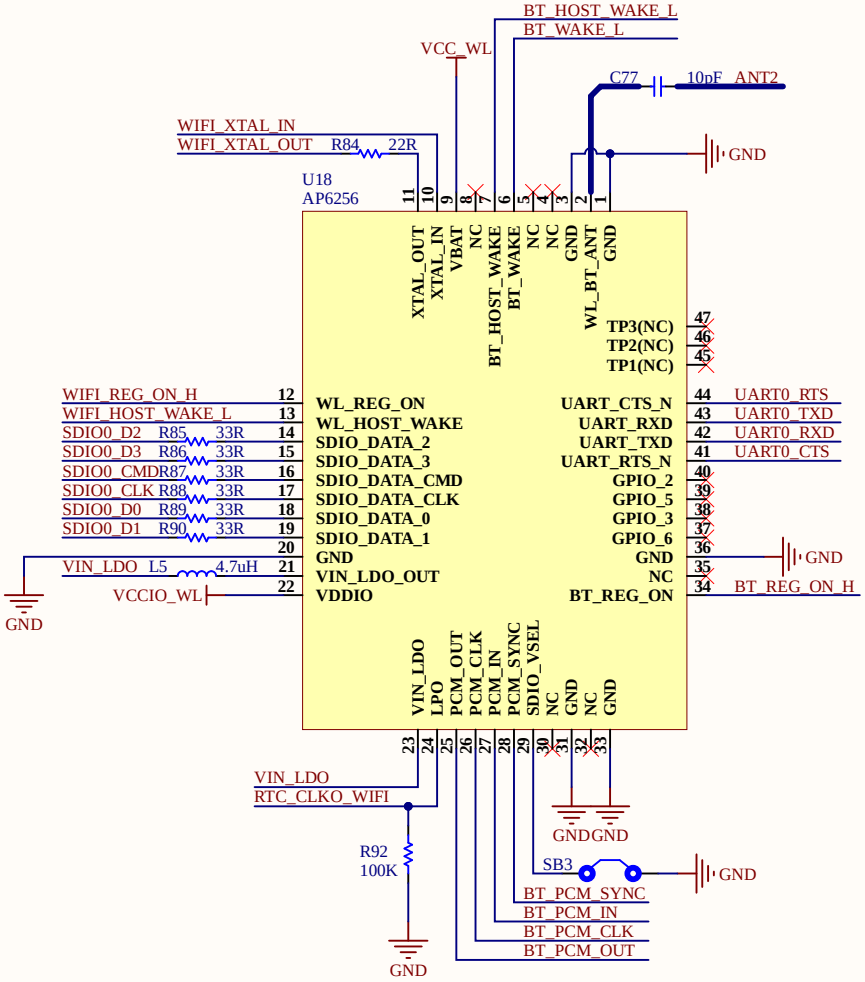
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WIFI_BLUETOOTH_INTERFACE

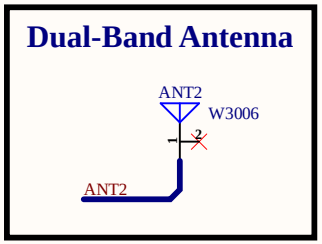
Crystal



Dual Band WiFi



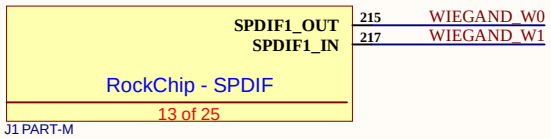
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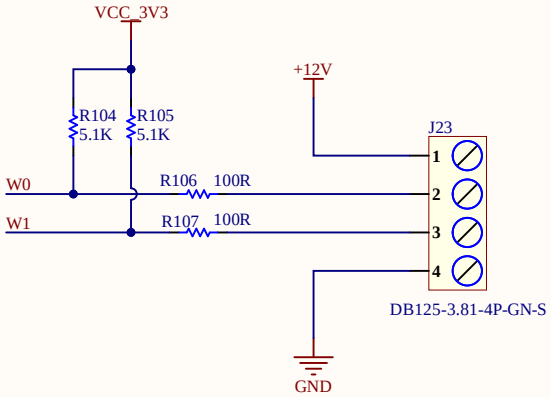


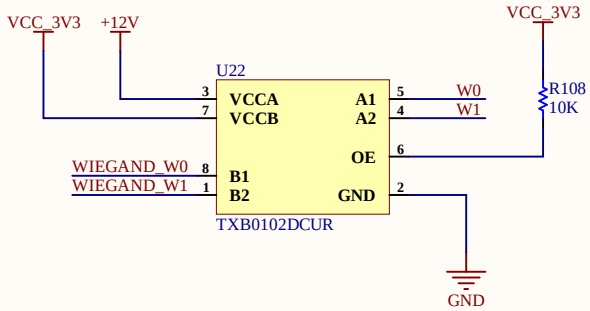
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WIEGAND_INTERFACE



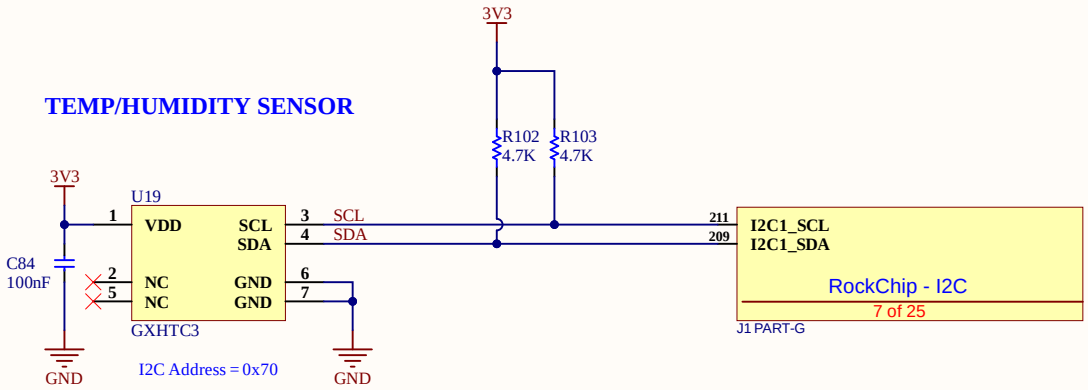




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TEMP/HUMIDITY SENSOR_INTERFACE



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